Akash Arvind Kore

Raleigh, NC | (919) 559-4937 | aakashkore@gmail.com | linkedin.com/in/akashkore | github.com/TheSelfSa

Work Authorization: F-1 STEM OPT | Open to Relocation

Summary

Machine Learning Engineer with **4+ years of experience** specializing in **LLM fine-tuning (LoRA/QLoRA), Retrieval-Augmented Generation (RAG), agentic workflows, and foundation model evaluation**. Shipped production AI systems delivering a **90% reduction in ML review time** and **35% inference cost savings**: a PyPI LLM evaluation framework (GPT-4o, 85-criteria scoring), a multi-LLM routing layer (OpenAI, Gemini, Anthropic, Groq), and an open-source RAG platform. M.S. Computer Science, NC State University.

Experience

Machine Learning Engineer — Aug. 2024 – Present
Ready Tensor, Inc. — *San Diego, CA (Remote)*

- Built and shipped **readytensor** (open-source PyPI package): an **LLM evaluation framework** scoring AI/ML repositories across 85 criteria via GPT-4o with seeded, deterministic inference, cutting ML review time by **90%**.
- Designed a **multi-LLM routing layer** unifying OpenAI API, Gemini, Groq, Anthropic, and OpenRouter with concurrent batch processing; integrated **LangChain** for **agentic workflow** orchestration, reducing inference costs by **35%**.
- Built production **MLOps pipelines** (Docker, AWS Lambda, GitHub Actions CI/CD) achieving **99.9% uptime**; executed zero-downtime database migrations and established IaC standards adopted team-wide.

Machine Learning Intern — May 2024 – Aug. 2024
Japan Atomic Energy Agency (JAEA) – Japan's national nuclear research lab — *Remote*

- Designed a **deep learning** pipeline for nuclear reactor bubble detection; selected **Pix2Pix GAN** (PyTorch) for sim-to-real **domain adaptation** under limited labeled-data constraints, achieving **90% object detection accuracy**, **30% improved generalization**.
- Curated a 50k-image labeled dataset; applied **transfer learning** fine-tuning to eliminate manual labeling; containerized the **real-time inference pipeline** (Docker, AWS Lambda) achieving 15 FPS throughput and **95% overhead reduction**.

Software Engineer Associate — Sep. 2021 – Jun. 2023
Siemens Digital Industries Software — *Pune, India*

- Engineered Teamcenter on Cloud (TcX) on **AWS (EC2, S3, RDS, Lambda)** for **500+ global enterprise customers**, supporting distributed data pipelines and telemetry; automated CI/CD via CloudFormation, cutting deployment setup time by **30%**.
- Developed and refactored **C++ server-side** components using OOP principles, reducing bug resolution time by **25%**; resolved **40+ critical production incidents** across Python, C++, and Bash in Agile sprints.

Research Intern — May 2020 – Jul. 2020
TATA Research, Development and Design Center — *Pune, India*

- Built an **anomaly detection** library (Python, Scikit-learn, NetworkX; OddBall graph-based ML) on healthcare datasets, improving accuracy by **15%**; deployed as a private **PyPI package** adopted by **5+ hospital partners** with **98% reliability**.

Projects

B.I.A.S.E.D. – RAG-Powered AI Decision Intelligence Platform github.com/TheSelfSa/biased — *Sep 2024 – Feb 2025*
FastAPI, Next.js, pgvector, Ollama, Docker, Better Auth | Open-source Apache-2.0

- RAG investigation engine that retrieves evidence from **pgvector** (vector database), routes queries across local Ollama or cloud LLMs via **agentic workflows**, and returns structured responses with citations and confidence traces.
- Built on a pnpm monorepo (Next.js PWA + FastAPI backend) with **RBAC**, Docker Compose deployment, and a scenario planning engine; released open-source under Apache-2.0.

End-to-End LLM Fine-Tuning Pipeline — *Jan 2025 – Apr 2025*
PyTorch, Hugging Face Transformers, LoRA/QLoRA (PEFT), vLLM, Weights & Biases, Ray Tune

- Fine-tuned **LLaMA-3** with **LoRA/QLoRA (PEFT)** on domain-specific conversational data; optimized hyperparameters with **Ray Tune** and tracked experiments via **Weights & Biases**; evaluated against baseline using ROUGE-L and BERTScore.
- Deployed quantized **model inference endpoints** via **vLLM** and FastAPI, achieving a **40% reduction in memory footprint**; serves as a production template for domain-specific **foundation model** adaptation and inference optimization.

Technical Skills

Languages: Python, JavaScript, TypeScript, Java, C++, SQL, Bash

LLMs & GenAI: GPT-4o, LLaMA-3, Claude (Anthropic), Gemini, OpenAI API, LLM Fine-Tuning (LoRA/QLoRA), PEFT, RAG, Retrieval-Augmented Generation, Prompt Engineering, Model Evaluation, Agentic Workflows, LangChain, vLLM, Foundation Models, Inference Optimization

AI / ML: PyTorch, TensorFlow, Scikit-Learn, Hugging Face Transformers, Deep Learning, NLP, Computer Vision, Distributed Training, A/B Testing, ONNX, TensorRT, Ray

MLOps & Cloud: MLflow, Weights & Biases, AWS (EC2, S3, RDS, Lambda, SageMaker), Vertex AI, Docker, Kubernetes, Terraform, GitHub Actions CI/CD

Data & Vectors: Vector Databases, pgvector, Pinecone, ChromaDB, FAISS, Apache Spark, Kafka, Airflow, PostgreSQL, MongoDB

Web & APIs: FastAPI, Django, Next.js, React.js, REST APIs, GraphQL, Pydantic, TailwindCSS

Education

Master of Science, Computer Science — Aug. 2023 – May 2025
North Carolina State University — *Raleigh, NC*

Bachelor of Technology, Computer Science — Aug. 2017 – Jul. 2021
Walchand College of Engineering — *Sangli, MH, India*